MACHINE LEARNING SECURITY: PROJECT 1

Evaluating robust accuracy of RobustBench models across different perturbations.

https://github.com/RDbtx/MLSECproject/tree/main

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Introduction

RobustBench is a standardized benchmark and open-source library for evaluating and compare adversarial robustness of image classifiers.

Our goal is **re-evaluating robust accuracy** of five CIFAR-10 (ℓ_∞) models to check the consistency of the results according to the ranking proposed by the official RobustBench leaderboard.

This process is done by using **AutoAttack suite** across different values of the perturbation ϵ.

Choosing the Models

We took five different models from RobustBench's **Model Zoo**. Each model reaches robustness by implementing different techniques. These are the chosen models:

Model (CIFAR-10 (ℓ_∞))	**Robust acc.**	**Robustness approach (summary)**
Peng2023Robust	71.07%	Generalizable architecture design principles: (1) choose an optimal depth/width range, (2) prefer **convolutional stem** over patchify stem, (3) **residual block** design, (4) **non-parametric** activation functions.
Rebuffi2021Fixing_70_16_cutmix_ddpm	64.20%	Combines **heuristics-driven** augmentations (Cutout, CutMix) with **data-driven** augmentation to increase diversity.
Wang2020Improving	56.29%	Improves robustness by **revisiting misclassified examples**; explicitly differentiates **misclassified vs correctly classified** training samples.
Rice2020Overfitting	53.42%	Improves robustness via **careful adversarial training** plus **early stopping** using precise checkpoints.
Chen2024Data_WRN_34_20	58.09%	**Classical adversarial training** + **data filtering** to reduce adversarial training examples, speeding training without hurting overall performance.

AutoAttack

We run the **AutoAttack module** to test the robustness of the models with two of the default proposed attacks.

They are:

→**APGD-CE** (Auto-PGD with Cross-Entropy loss)
→**APGD-T** (Auto-PGD Targeted)

```
if mode == "fast untargeted":
    print("Fast mode selected [APGD-CE only]!")
    adversary = AutoAttack(model, norm="Linf", eps=eps, version="custom", device=device, verbose=verbose)
    adversary.attacks_to_run = ["apgd-ce"]

elif mode == "fast targeted":
    print("Fast mode selected [APGD-T only]!")
    adversary = AutoAttack(model, norm="Linf", eps=eps, version="custom", device=device, verbose=verbose)
    adversary.attacks_to_run = ["apgd-t"]
```

Implementation

- This pipeline does not need to set specific attack parameters, just testing it each time with a different value of perturbation **ε**.
- We made also our code **scalable** and **modular** to test the other types of proposed AutoAttacks with different models.

1. **Load CIFAR-10 test set** and pick a **reproducible random subset** (≈100–200 samples).

2. Define an **epsilon grid** (default:1/255,4/255,8/255,12/255,16/2551/255).

3. For each **model × ε**, run **AutoAttack** and compute **robust accuracy** (+ runtime/config metadata).

4. **Save results** as **full JSON** + **CSV table** (models × ε).

5. From the CSV, generate **plots**: robust-accuracy curves, **rank vs ε**, and a **heatmap** ordered by average rank.

Difficulties

•Running **AutoAttack** across **5 models × multiple ε values × CIFAR-10 samples** resulted in **very long execution times**.

•Full AutoAttack (APGD-CE, APGD-DLR, FAB, Square) is **robust but slow**, especially when repeated the ε-grid that we designed.

•To make the pipeline usable while preserving meaningful robustness signals, we introduced **two fast attack modes** in the codebase (labelled as fast untargeted and fast targeted).

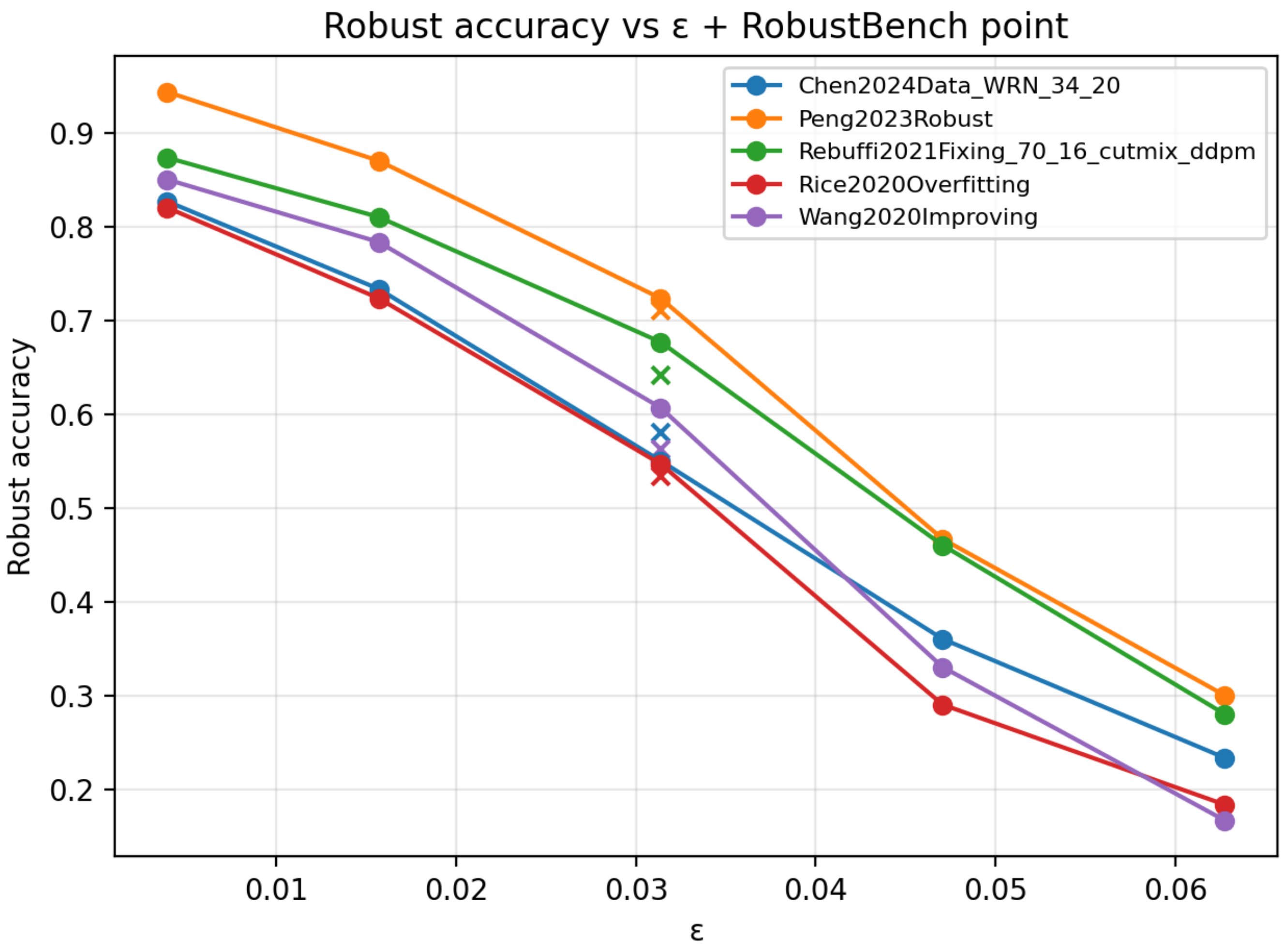
•We accepted the tradeoff:

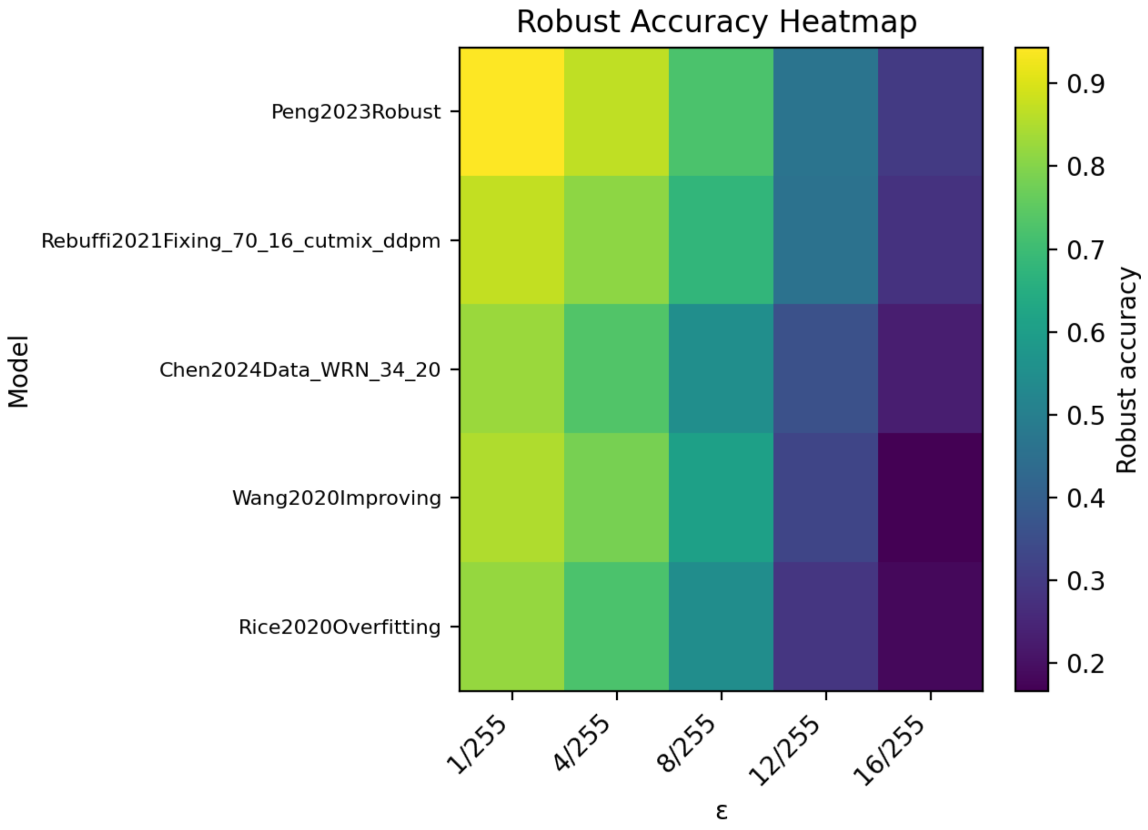
→ **Much faster evaluation**;

→ Slightly less exhaustive than full AutoAttack;

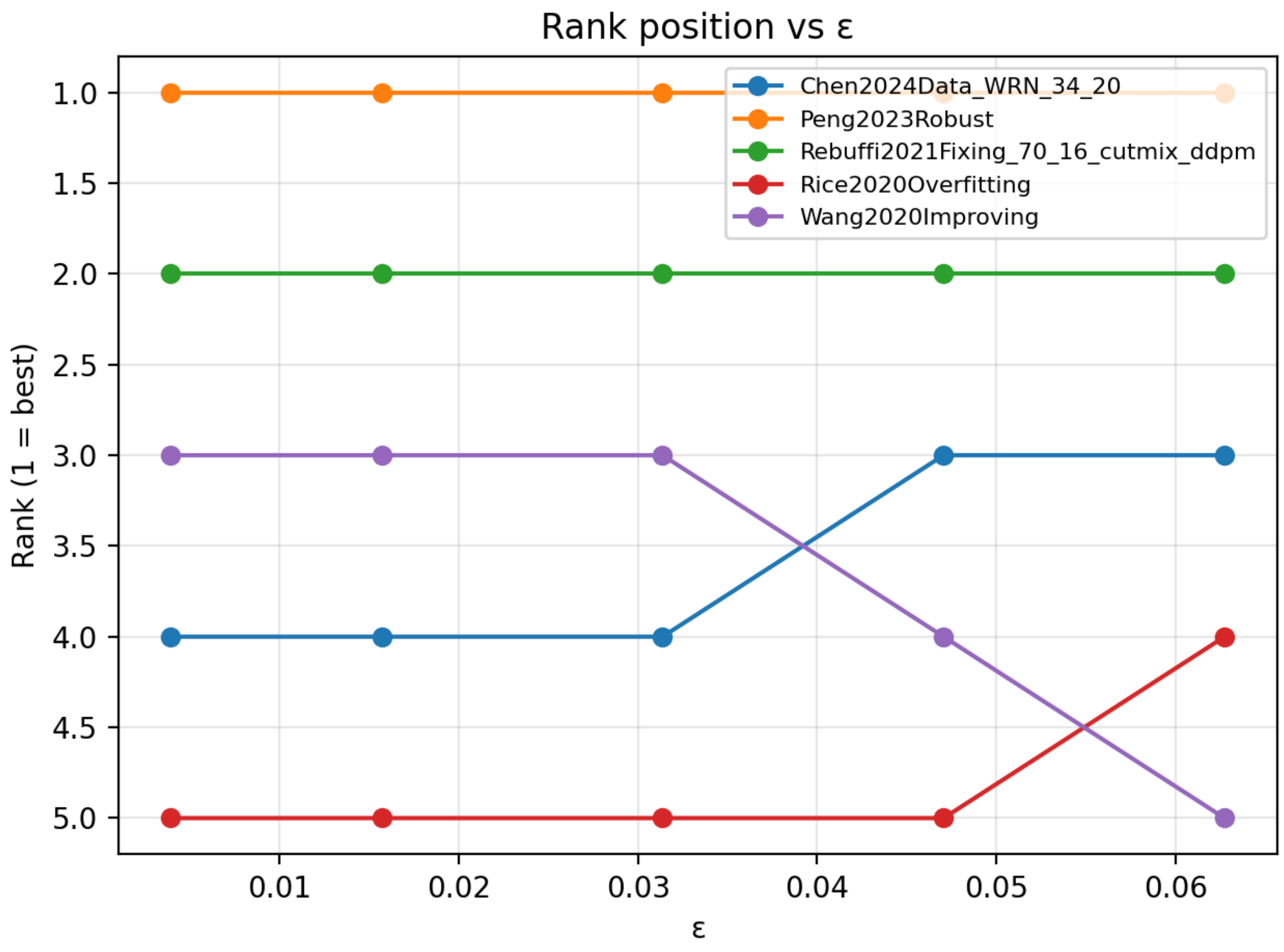
→Still consistent for **relative robustness ranking** and trend analysis.

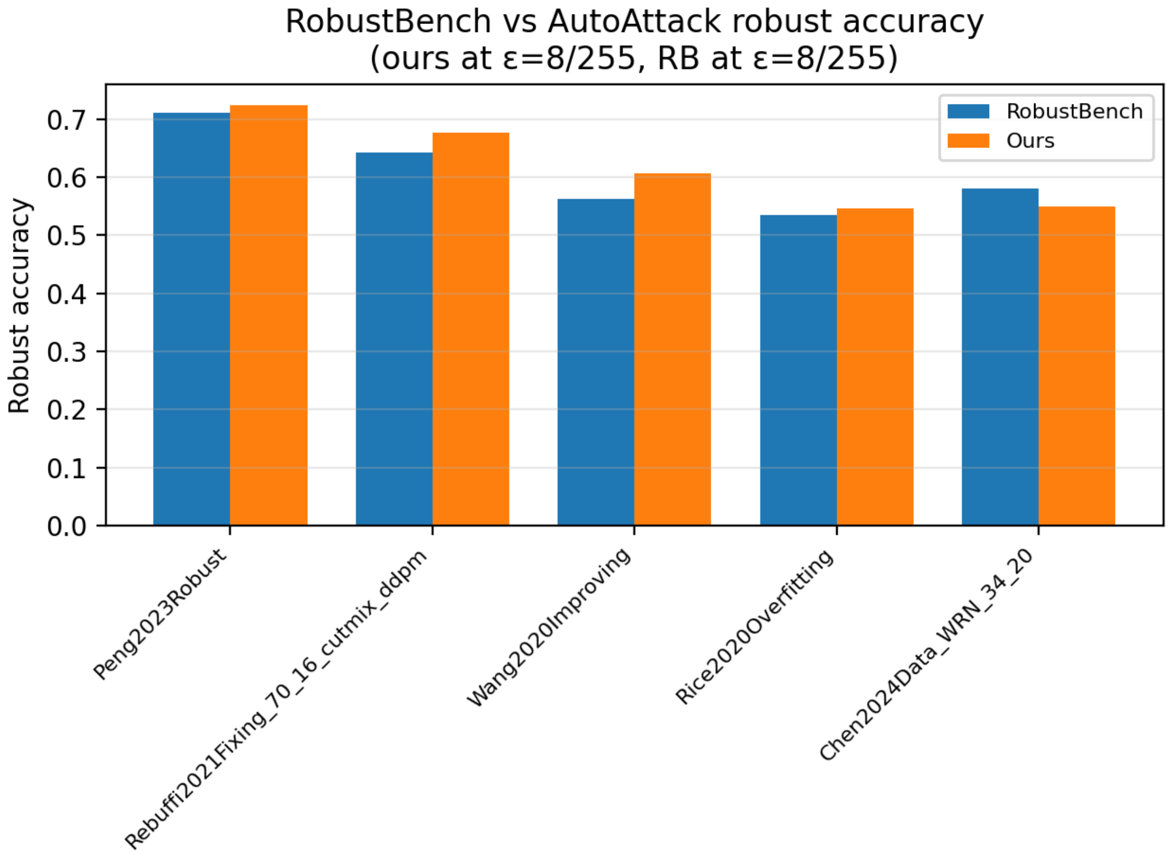
APGD-CE/APGD-T Results





APGD-CE/APGD-T Results





Conclusions

- **Robust accuracy decreases monotonically as ε increases** for all models → robustness should be assessed **as a robustness–ε curve**, not from a single ε point.
- **Rankings are mostly stable up to ε = 8/255**; beyond that, **especially weaker models show larger ranking fluctuations**.
- At **ε = 8/255**, our **AutoAttack (APGD-CE/APGD-T) recomputation is consistently lower** than RobustBench → RobustBench appears **optimistic in absolute robustness**. This gap may also be affected by our choiche of not running the full AutoAttack suite (**FAB-T** and **Square**) due to hardware/time constraints.
- **For our selected models, RobustBench's single-point ranking at ε = 8/255 does not generalize across ε**: robustness curves expose different degradation rates, and **Wang2020** in particular shifts from near-top to **worst** at higher ε. Therefore, defenses should be compared using **robustness-vs-ε curves under strong attacks**, not one ε value.

THANK YOU FOR THE ATTENTION !

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